Soil CO_2 and N_2O emissions in Sahelian Silvopastoral systems: spatial distribution and annual budget estimation

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Sahelian Silvo Pastoral Systems (SPS)





SPS are a mix of trees, pastures and livestock in the same area.

- Trees deliver important ecosystem services (cooling benefits, potential to sequester carbon and offset greenhouse gas (GHG) emissions…),
- Pastures feed livestock and livestock provides food and supports livelihoods for many people in the Sahel

SPS are an attractive nature-based climate solution, **but** the impact of livestock on carbon cycle and GHG fluxes in Sahelian SPS is poorly studied.

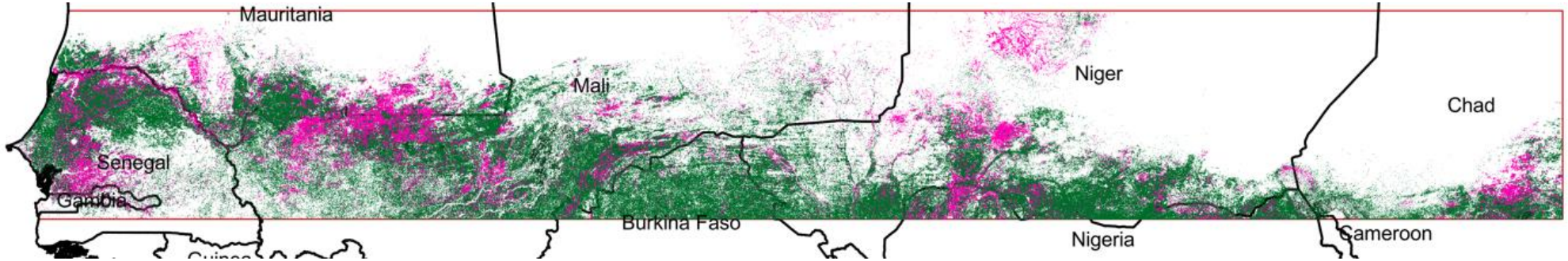
Modeling studies offer insight into the processes and spatial/temporal variability of GHG fluxes and provide estimates from local to regional scales

Research Questions

- How soil CO_2 and N_2O emissions varied from 2010 to 2021 in Sahelian SPS?
- What are the spatial patterns of CO_2 and N_2O emissions from soils in the Sahelian SPS?
- What environmental factors control the spatial changes in soil CO_2 and N_2O emissions?

Where?

Sahelian silvopastoral systems (SPS) of West-Africa range from longitude 18°W to 20°E and latitude 13°N to 18°N, which cover approximately 40% (≈ 892,353 km^2) of the Sahelian band

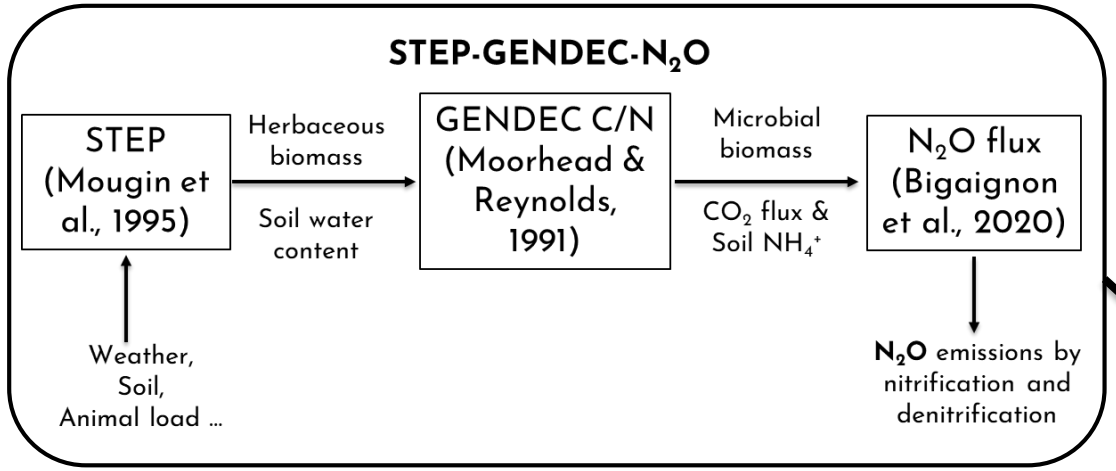


FAO GLC-SHARE dataset

How?

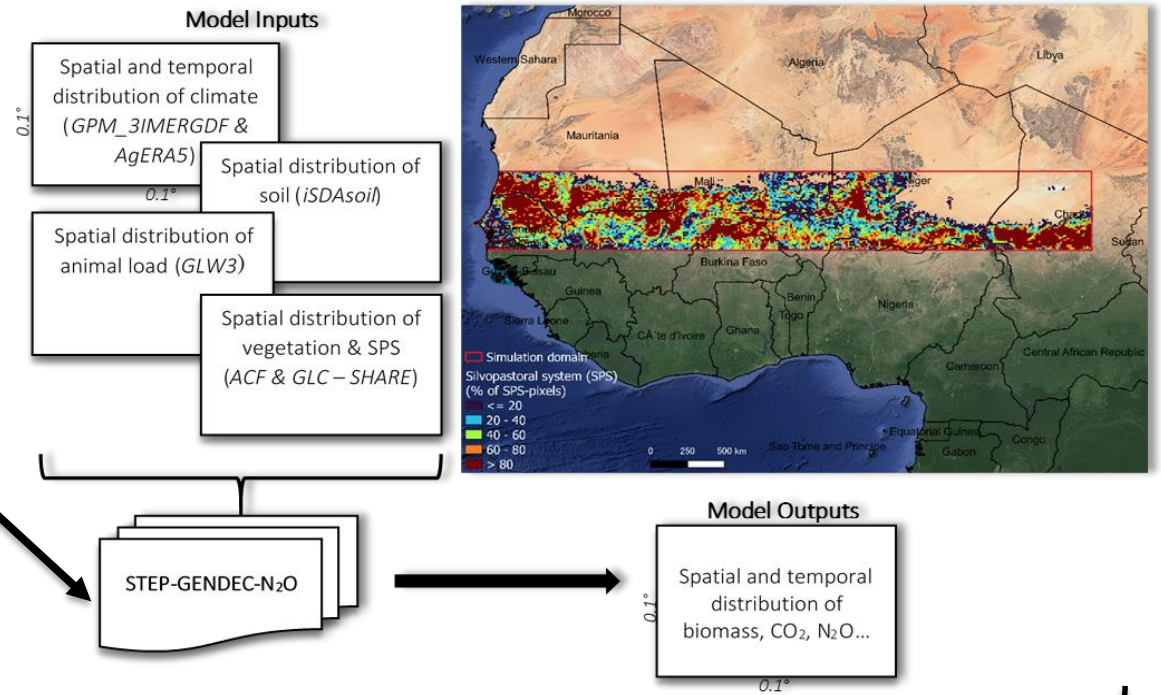
- **STEP-GENDEC-N_2O:** process-based model coupling water budget, vegetation growth, biogeochemistry and gaseous emissions.

STEP-GENDEC-N_2O validated locally → **Up-scaled to all SPS of the Sahel region**



Previous studies in sites representative of the Sahelian SPS :
Agbohessou et al., 2023 (under review); Bigaignon et al., (2020); Delon et al., (2019); Delon et al., (2015)

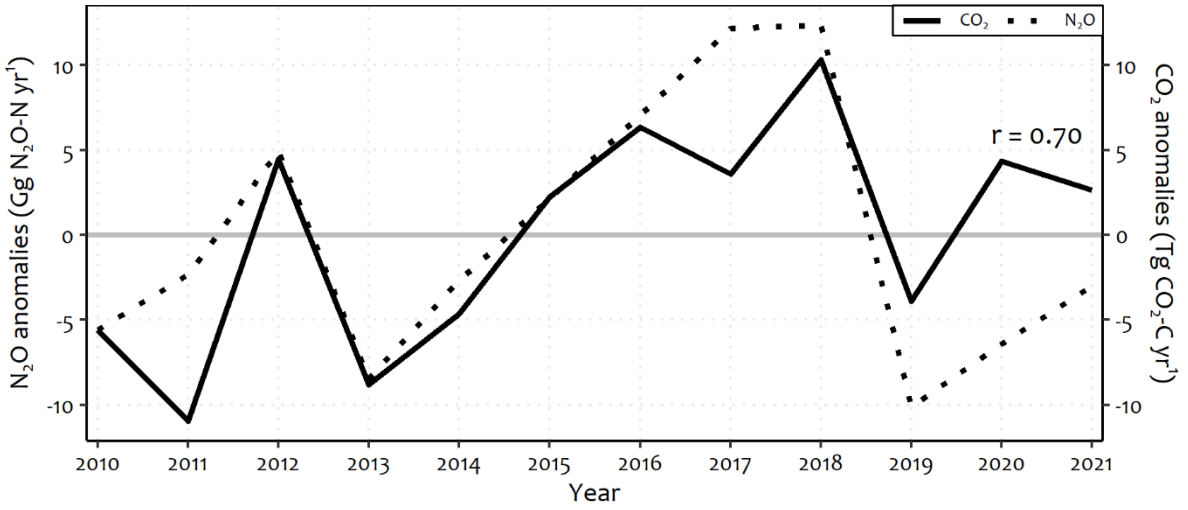
1D : local scale

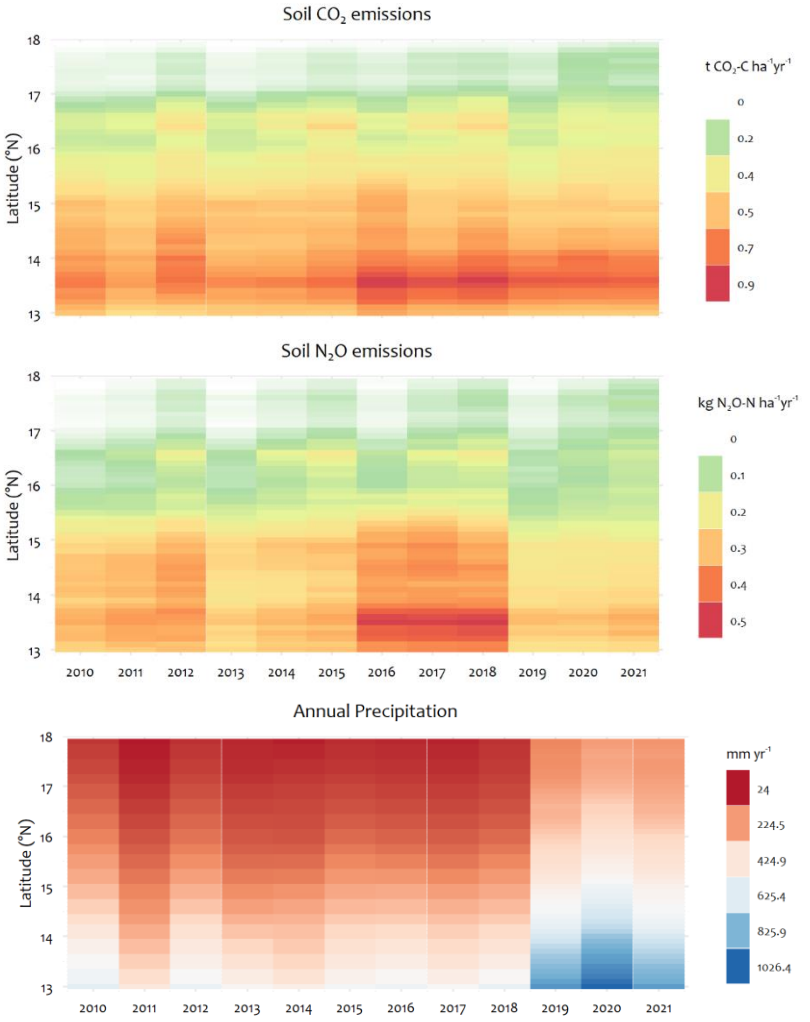


2D : regional scale

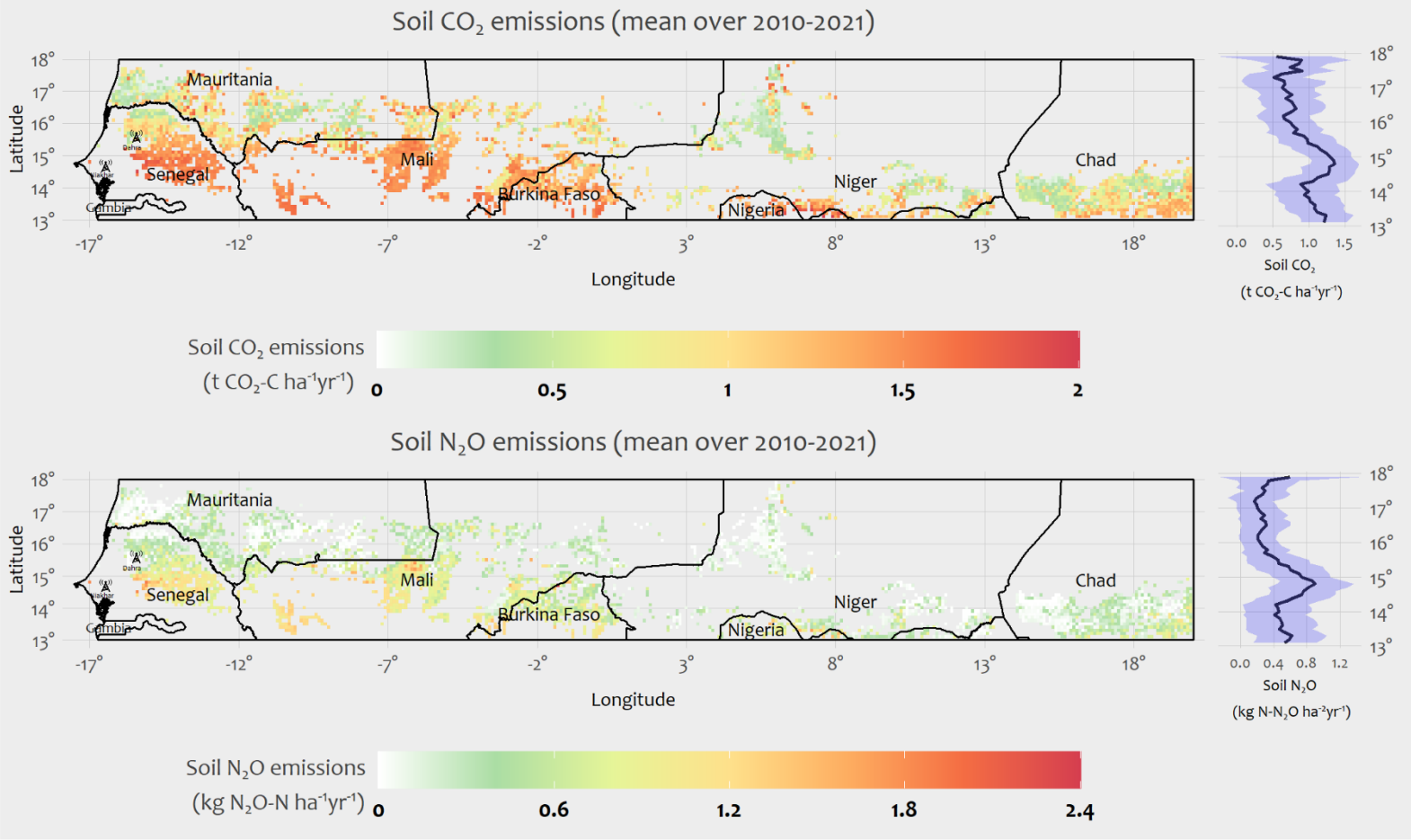
How soil CO_2 and N_2O emissions varied from 2010 to 2021 in SPS?

- Soil CO_2 emissions include the respiration of soil organisms (microbial respiration) and plant roots
- We calculated the proportion of SPS area pixels within each 0.1° x 0.1° simulation grid cell, and used these values to weight the model outputs for each grid cell.



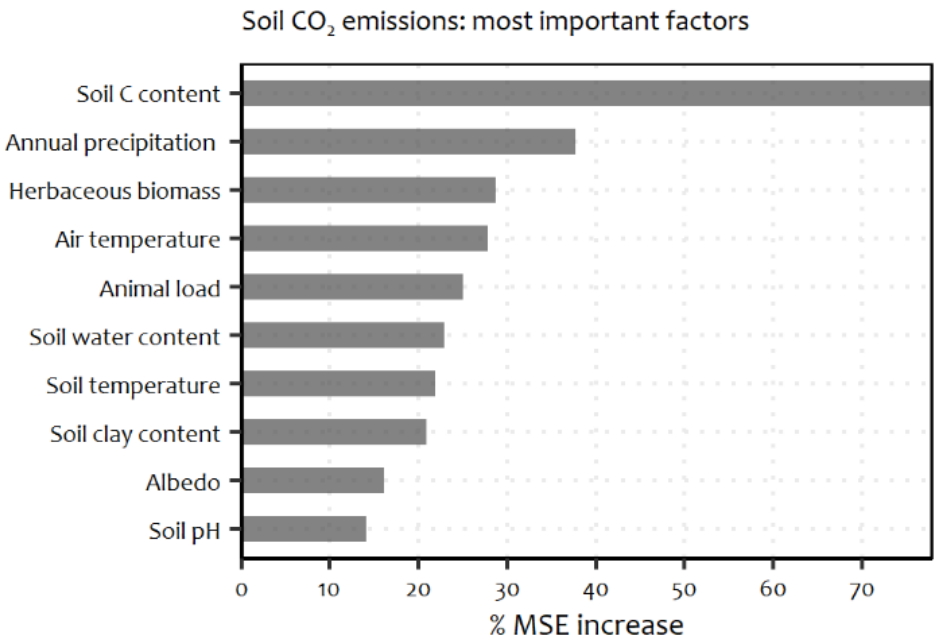


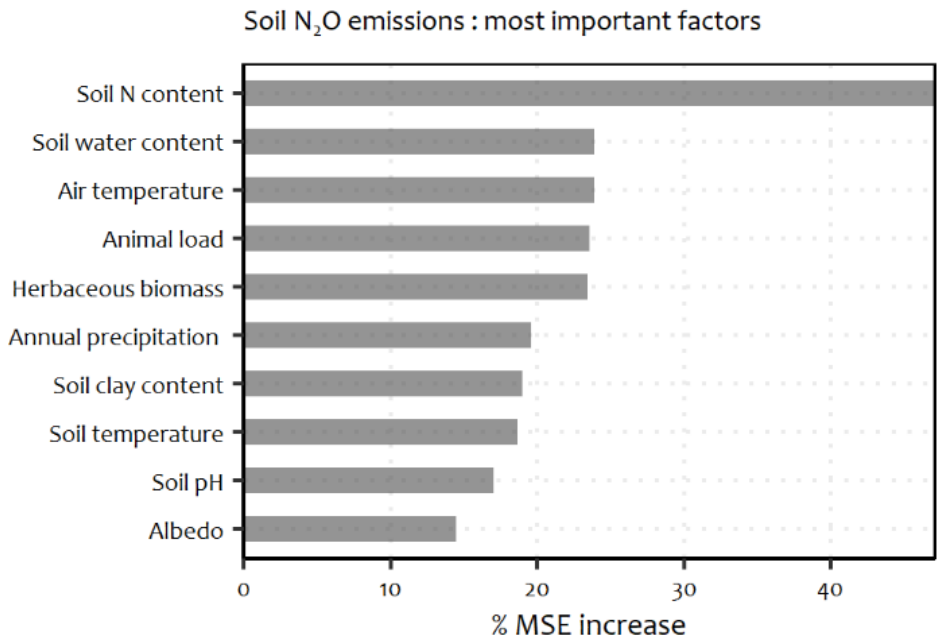
How soil CO_2 and N_2O emissions varied spatially in SPS?



- Only pixels dominated by SPS (>80%) are displayed on the above emissions maps

What factors control the spatial changes in soil CO_2 and N_2O emissions?





- We used the Random Forest algorithm to determine the key factors that affect the spatial patterns of soil CO_2/N_2O emissions.

Take-home message

- There is a latitudinal gradient of CO_2 and N_2O emissions, with lower emissions in the north and higher emissions around 15°N (2010-2021).

- Significant upward trends in soil CO_2 and N_2O emissions were observed between 2013 and 2018 in the Sahelian SPS.

- The distribution of CO_2 and N_2O emissions is heterogeneous, and is primarily controlled by factors such as substrate and water availability, herbaceous mass, air temperature and animal load.

Thank you for your kind attention

References :

- Mougin, E., Lo Seena, D., Rambal, S., Gaston, A., Hiernaux, P., 1995. A regional Sahelian grassland model to be coupled with multispectral satellite data. I: Model description and validation. Remote Sensing of Environment 52, 181–193. https://doi.org/10.1016/0034-4257(94)00126-8
- Moorhead, D.L., Reynolds, J.F., 1991. A general model of litter decomposition in the northern Chihuahuan Desert. Ecological Modelling 56, 197–219. https://doi.org/10.1016/0304-3800(91)90200-K
- Bigaignon, L., Delon, C., Ndiaye, O., Galy-Lacaux, C., Serça, D., Guérin, F., Tallec, T., Merbold, L., Tagesson, T., Fensholt, R., André, S., Galliau, S., 2020. Understanding N_2O Emissions in African Ecosystems: Assessments from a Semi-Arid Savanna Grassland in Senegal and Sub-Tropical Agricultural Fields in Kenya 26.
- Agbohessou, Y.F., Delon, C., Mougin, E., Grippa, M., Tagesson, T., Diedhiou, M., Ba, S., Ngom, D., Vezy, R., Ndiaye, O., Roupsard, O., 2023. Is the Impact of Livestock on Greenhouse Gas Emissions Offset by the Presence of Trees in a Sahelian Silvopastoral System? Available at SSRN 4357164 (Under review).
- Delon, C., Galy-Lacaux, C., Serça, D., Personne, E., Mougin, E., Adon, M., Dantec, V.L., Loubet, B., Fensholt, R., Tagesson, T., 2019. Modelling land–atmosphere daily exchanges of NO, NH_3, and CO_2 in a semi-arid grazed ecosystem in Senegal 29
- Delon, C., Mougin, E., Serça, D., Grippa, M., Hiernaux, P., Diawara, M., Galy-Lacaux, C., Kergoat, L., 2015. Modelling the effect of soil moisture and organic matter degradation on biogenic NO emissions from soils in Sahel rangeland (Mali). Biogeosciences 12, 3253–3272.

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